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# Physical and Mechanical Properties of Blends Based on Poly (dl-lactide), Poly (l-lactide-glycolide) and Poly (ϵ-caprolactone)

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# Physical and Mechanical Properties of Blends Based on Poly (dl-lactide), Poly (l-lactide-glycolide) and Poly (ε-caprolactone)

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**Bioresorbable materials are extensively used for a wide range of biomedical applications. In this study, the common industrial processes of compression moulding and solvent casting were utilised for initial preparation of thin film blends based on Poly (dl-lactide), Poly (l-lactide-glycolide) and Poly (ε-caprolactone). Attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR), differential scanning calorimetry (DSC), phosphate buffered saline adsorption, tensile testing and contact angle measurement were used as a means of investigating the physical and mechanical properties of the blends.**

**Keywords** Biodegradable; Bioresorbable; Blends; DSC; FTIR; Tensile testing

## INTRODUCTION

Resorbable and degradable polymers have been extensively studied throughout the last few decades. A resorbable material can be broken down and the degradation byproducts eliminated from the body[1]. In polymers, the resorbability occurs mainly through enzymatic or non-enzymatic hydrolysis followed by the metabolism or excretion of the degradation products. The use of the term resorbable is limited to biomedical materials. When degradable polymers are used elsewhere, they should be denoted degradable or biodegradable[2]. Aliphatic polyesters have found prominence among synthetic resorbable polymers for implants, the most extensively studied amongst this family of polymers are Poly (l-lactide), Poly (glycolide), Poly (ε-caprolactone) and copolymers based on l/dl-lactide, glycolide, trimethyl carbonate and ε-caprolactone[3]. The benefits of using resorbable or degradable materials as biomedical materials are evident as biomedical materials implanted *in vivo* degrade after serving their purpose. This has been utilised in the preparation of controlled drug release systems, in orthopaedic implants and sutures[4].

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In many cases, improvements in physical and mechanical properties can be imparted more rapidly and cost-effectively by mixing available polymers (using existing processing equipment) rather than developing new chemistry. Currently, blending aims at securing sets of specific properties required for an envisaged application[5]. The final properties of such blends depend on the chemical structure of the original components, the mixing ratio of the constituent polymers, the interaction between the components and the processing steps to which they have been subjected[6,7]. In this study, blends based on Poly (dl-lactide) (PDLLA), Poly (l-lactide-glycolide) (PLGA) and Poly (ε-caprolactone) (PCL) were fabricated using compressing moulding and solvent casting techniques, in an attempt to produce polymeric films with a board range of physical and mechanical properties. These blends were then analysed using ATR-FTIR, DSC, phosphate buffered saline adsorption, mechanical analysis and contact angle measurement.

## EXPERIMENTAL

### Materials

The Poly (ε-caprolactone) (brand name Tone Polymer, P-676 TM) was supplied by the Dow Chemical Company; the Poly (l-lactide-glycolide) copolymer with a molar ratio of 83/17 (brand name Purasorb PLG) was obtained from Purac Biochem Bv Gorinchem; while the Poly (dl-lactic acid) (brand name Galastic, PABR-L-68) was obtained from Galactic Laboratories (a division of Brussels Biotech), all in granular form. The solvents used included dichloromethane, tetrahydrofuran, ethyl acetate, chloroform, acetone and methanol, all of which were HPLC grade (supplied by Aldrich).

### Solvent Casting Procedure

The solvents investigated for dissolving the base polymers (PLGA, PCL and PDLLA) were dichloromethane, tetrahydrofuran, ethyl acetate and chloroform. Different ratios of polymer to solvent were initially investigated, with the optimum ratio found to be 5 g polymer to 50 ml solvent. It was found that chloroform was the most efficient solvent at room temperature for the base polymers, thus this solvent was chosen for all solvent cast blend preparations. A two-step procedure was developed for the solvent casting of the thin films. The first step involved dissolving the polymer or polymer mixtures; 5 g of the polymer or polymer mixtures were weighed and placed into a glass beaker, 50 ml of chloroform was added and the polymer/solvent mixture was agitated using a magnetic stirrer for a period of between 2–4 hr until the polymers were fully dissolved. The second step involved removal of the solvent and formation of the film; the polymer/solvent solution was poured into a flat glass container, the solvent was removed via evaporation at room temperature under a fume hood for a period of 24 hr and for 12 hr in an oven at a temperature of 40°C. The mixing ratio of the thin film polymer blends prepared by both the solution casting and compression moulding techniques are presented in Table 1. Note that for all experimental works in this study, weights were measured using a Sartorius scales capable of being read to five decimal places.

## Compression Moulding Procedure

During initial trials, it was found that the compression moulding technique did not provide adequate mixing of the blends. As a consequence, a pre-mixing step was deemed necessary prior to the compression moulding process. The technique identified as most suitable was precipitation mixing, which involved the following steps; 5 g of the polymer or polymer mixture was weighed, placed into a flat-bottomed flask containing 50 ml of acetone, the flask was placed on top of a heating mantle magnetic stirrer apparatus with a reflux condenser attached and the mixture was heated to 35°C. The polymer/solvent mixture was agitated until the polymers had dissolved. The heated solution was poured into a glass petri dish and 20 ml of methanol was added (addition of the methanol and cooling of the mixture forces the polymers out of solution). Finally, the solvents were removed via evaporation at room temperature in a fume hood for a period of 24 hr.

A Daniels compression press, flat mould and a pressure of 1000 psi was used for the fabrication of the polymer films. Following processing trials, a sample weight of 1 g, a moulding cycle time of 1 min and a temperature of 170°C were chosen, as these parameters produced thin films with optimum properties. Also Teflon sheets were used to avoid polymer adhesion to the mould, while all samples were pre-heated for a time period of 1 min on a hotplate set at a temperature of 100°C, prior to the compression moulding cycle. The final steps involved taking the sample from the mould, cooling at room temperature for a number of minutes, and removal of the thin film from the Teflon sheets. The thickness of the thin films produced was recorded using a digital micrometer.

TABLE 1
Mixing ratio of thin film polymer blends based on PDLLA, PLGA and PCL

| Polymer blend (Percentage polymer) | PDLLA (%) | PLGA (%) | PCL (%) |
|---|---|---|---|
| PDLLA20/PCL80 | 20 | – | 80 |
| PDLLA50/PCL50 | 50 | – | 50 |
| PDLLA80/PCL20 | 80 | – | 20 |
| PLGA20/PCL80 | – | 20 | 80 |
| PLGA50/PCL50 | – | 50 | 50 |
| PLGA80/PCL20 | – | 80 | 20 |
| PDLLA80/PLGA20 | 80 | 20 | – |
| PDLLA50/PLGA50 | 50 | 50 | – |
| PDLLA20/PLGA20 | 20 | 80 | – |

## Attenuated Total Reflectance Fourier Transform Infrared Spectroscopy

Attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR) was carried out on rectangular samples using a Nicolet Avator 360 FTIR, with a 32 scan per sample cycle.

## Differential Scanning Calorimetry

Differential scanning calorimetry (DSC) analysis was carried out using a Perkin-Elmer, Pyris 6 DSC. Samples of between 9.3–10.3 mg were prepared from the base polymers and polymer blends. All measurements were conducted in crimped non-hermetic aluminium pans with an empty crimped aluminium pan being used as the reference cell. To assess the melting points and miscibility of the polymer(s), the samples were heated from a temperature of 20°C to 200°C at a rate of 10°C per minute. For investigation of the crystalline interactions, samples were heated from a temperature of 20°C to 200°C at a rate of 10°C per minute and held isothermal for a 5-min period. The samples were then cooled down from 200°C to 20°C at a rate of 2°C per minute. All DSC tests were carried out under a 20 ml/min flow of nitrogen to prevent oxidation. High-purity indium was used to calibrate the DSC cell.

## Phosphate-Buffered Saline Adsorption

The testing procedure for the phosphate buffered saline (PBS) adsorption was based on ASTM international standard D570-81 standard test method for water adsorption of plastics. This test covers the determination of the relative rate of adsorption of water by plastics when immersed. Three samples of each batch were cut into 60 mm by 10 mm strips and placed in test-tubes and dried in an oven for 24 hr at a temperature of 50°C. Samples were immersed

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in PBS at a temperature of 50°C. The PBS was made using distilled water and PBS tablets (Sigma Aldrich). After regular time intervals (24, 48 and 120 hr), samples were taken from the test tubes, surface water was removed using blotting paper, the samples were weighed and re-immersed in PBS. The percentage weight was calculated using the following equation:

$$(W_a - W_i/W_i) \times 100$$

where $W_i$ is the initial weight of the sample and $W_a$ is the weight of the sample after the adsorption period.

### Mechanical Properties

The tensile testing procedure used was based on the ASTM international standard D882-02 standard test method for tensile properties of thin plastic sheeting. The samples were cut into tensile samples using a dumbbell cutter, which gave a sample test length of 20 mm and a width of 4 mm. Five samples were tested from each batch using a Lloyd Lrx Tensile tester, with Nexygen software. A cross-head speed of 20 mm/min was used during all tests. To reduce slippage and sample breakage in the jaw area, paper labels were applied to the samples.

### Surface Analysis

The surface properties of blends and base polymers were assessed using contact angle analysis. The contact angles of water on the thin film polymers were measured using a Rame-Hart Goniometer at room temperature (20°C). The drops were made using a Gilson Pipetman set to 0.1 ml.

## RESULTS AND DISCUSSION

### Fabrication of Polymer Blends

The first technique investigated for the formation of the thin film polymer blends was solvent casting. Solvent casting has been an established process for the preparation of polymeric films for decades[8]. One of the most important considerations in solvent casting is the choice of solvent. Initial trials were carried out using several commonly used solvents, e.g., dichloromethane, chloroform, tetrahydrofuran and ethyl acetate, in an attempt to determine the most suitable solvent to dissolve PLGA, PCL and PDLLA. Based on these trials, the most effective solvent proved to be chloroform, in agreement with previous experimental works[9–11]. It was found that when the blends of PLGA/PCL or PDLLA/PCL were mixed at the following blend ratios of 50:50 and 80:20, there were undulations formed on the surface of the solvent cast films. These surface undulations led to a polymer film with variable thickness. Therefore compression moulding was used in an attempt to produce consistent thin polymeric films at higher PLGA and PDLLA ratios.

Three of the most important processing parameters of compression moulding are heat, time and pressure[8]. To produce films of consistent thickness, experimental work was carried out to ascertain the optimum process temperature and cycle time. During the initial trials, it was found by visual examination of the blend films that there was little or no mixing of the polymers during the compression moulding process. Therefore, in order to produce consistent and homogenous thin film polymer blends, a pre-mixing step was deemed appropriate prior to compression moulding. An economical lab-scale mixing process was required that did not waste material, thus precipitation mixing was chosen. This is a form of solution blending that is carried out by dissolving the two components in a common solvent and precipitating out the blend by addition of a suitable precipitant[12]. During the precipitation mixing process, the polymers were first dissolved in acetone by heating to approximately 40°C.

Acetone was chosen as it partially dissolves PLGA, PDLLA and PCL at room temperature and requires only gentle heating to fully dissolve the polymers[10,13,14]. As the polymer/acetone mixture was allowed to cool, methanol was mixed with the polymer solution. The cooling of the mixture and addition of the methanol forced the polymers out of solution, which resulted in the polymer mix. The solvents were removed by evaporation at room temperature in a vacuum hood for a period of 24 hr. Different process settings for heat and time with the same mould pressure (1000 psi) were trialled using the blends. Finally cycle times of 1 minute and a mould temperature of 170°C were decided upon as films with consistent thickness were produced using these parameters. The thin film samples produced using the compression moulding technique had an approximate thickness of 1.2 mm ± 0.3 mm. To gain knowledge of the miscibility and physical properties of the thin film polymer blends at diverse blend percentages, a variety of mixing ratios were investigated as illustrated in Table 1. As a result of the inconsistent nature of the film thickness produced using the solvent casting process, compression moulded samples were chosen for all subsequent analysis in this study.

### Infrared Spectroscopy of Thin Film Polymer Blends

Infrared spectroscopy (IR) is an invaluable analytical tool for both qualitative and quantitative analysis. The infrared absorbance spectrum of a given substance can be interpreted through known frequency IR bands, which correspond to certain functional groups each having a unique absorbance frequency band. In this study, the IR peaks for the compression moulded PCL, PDLLA and PLGA samples were in good agreement with the characteristic bands reported by Coleman and Zarian (1979), Garlotta (2001), Vey et al. (2008) and Choi et al. (2008) for these homopolymers[15–18]. The behaviour of polymer blends depends, in

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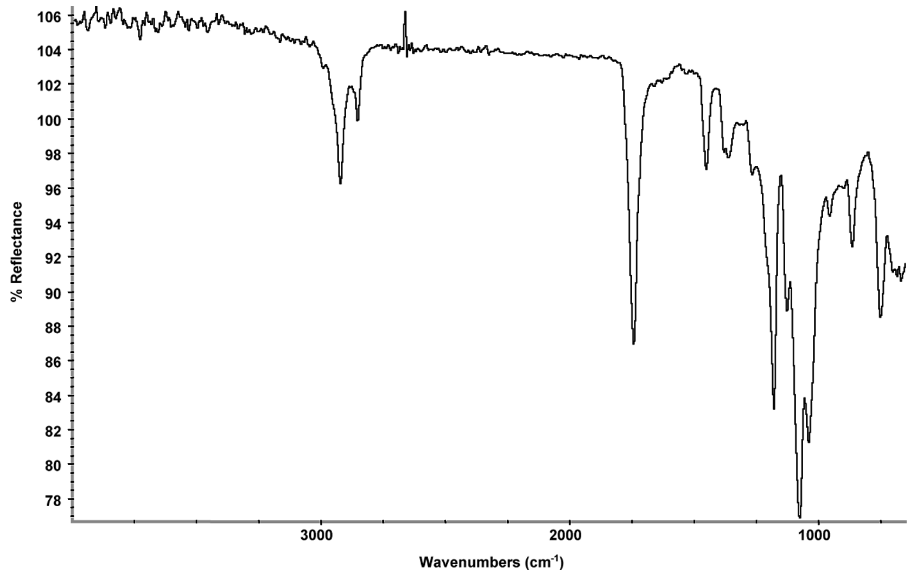


FIG. 1. Infrared spectrum of compression moulded 50%PDLLA/50%PCL thin film polymer blend.

general, on the mixing degree of the components and their mutual interaction, as well as on the individual properties of these components.

From an IR point of view, compatibility of a blend is defined in terms of the presence of a detectable ‘‘interaction’’ spectrum that arises when the spectrum of the blend is compared to the spectra of the two homopolymers. A representative infrared spectrum of one of the compression moulded thin film blends (PCL 50% PDLLA/50% PCL) is presented in Figure 1. Throughout this study, there were no peaks present in the IR spectra of the blends that could not be explained by the characteristic IR bands of the base polymers. This correlates with work carried out using FTIR by Chen et al. (2003) on blends based on PCL[6].

## Differential Scanning Calorimetry

Thermal analysis is very important for the evaluation of polymers and polymer blends. Differential scanning calorimetry (DSC) is a technique in which heat flow into a substance and a reference sample is measured as a function of temperature. DSC was used to analyse the effect that blending of the polymers had on the endothermic ($T_g$ and $T_m$) peaks of the constituent polymers and to assess what impact blending PDLLA and PLGA with PCL had on the crystallinity of the samples. The DSC thermographs of PDLLA, PLGA and 50%PDLLA/50%PLGA are shown in Figure 2. PLGA is a semi-crystalline polymer for which two peaks were exhibited: $T_g$ at 65.7°C and $T_m$ at 145°C. The results obtained fall within the range of values for $T_g$ (40–70°C) and $T_m$ (125–225°C) reported by Gilding and Reed (1979) for PLGA[19]. Values of $T_g$ and $T_m$ for PLGA have been found to be dependent on the monomer ratio of glycolide to l-lactide[19]. PDLLA is an amorphous polymer and exhibits only one peak, a $T_g$ at approximately 48.8°C. This result is within the range of values (47.2–51.6°C) obtained for the $T_g$ of PDLLA by Chen et al.[6] In the 50% blend of the two polymers, a single $T_g$ was evident at 54°C. As only one peak was observed for the blend, this would indicate a level of miscibility between the two polymers.

In Figure 3, the thermographs of the 80%PDLLA/20%PLGA, 50%PDLLA/50%PLGA and 20%PDLLA/80%PLGA blends are presented. The DSC thermographs show that there is single $T_g$ for each of the blends and as the percentage of PLGA in the blend increases there is a slight increase in the $T_g$ values. Phase separation of binary blends of PLA and PLGA is not as likely as with many other polymer combinations due to their similarity in

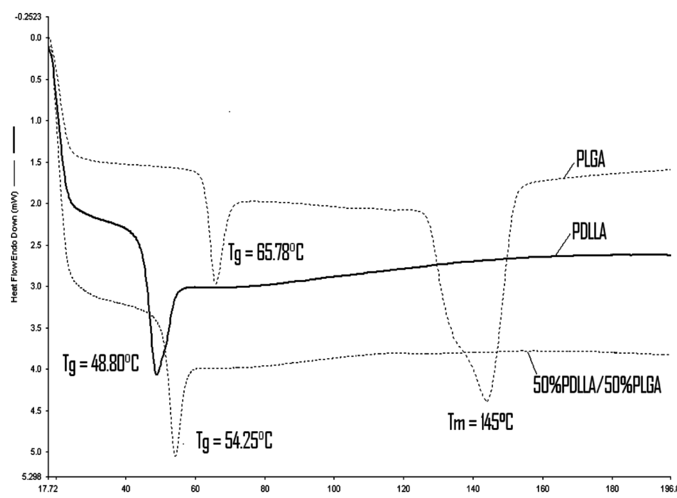


FIG. 2. DSC thermographs of PDLLA, PLGA and a 50%PDLLA/50%PLGA blend.

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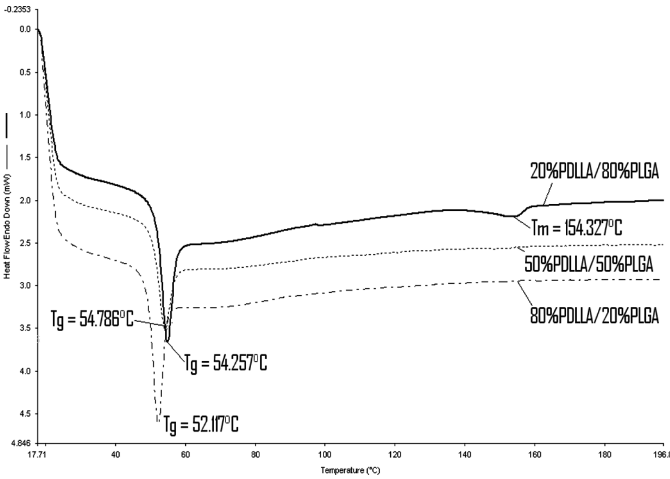


FIG. 3. DSC thermographs of 80%PDLLA/20%PLGA, 50%PDLLA/50%PLGA and 20%PDLLA/80%PLGA blends.

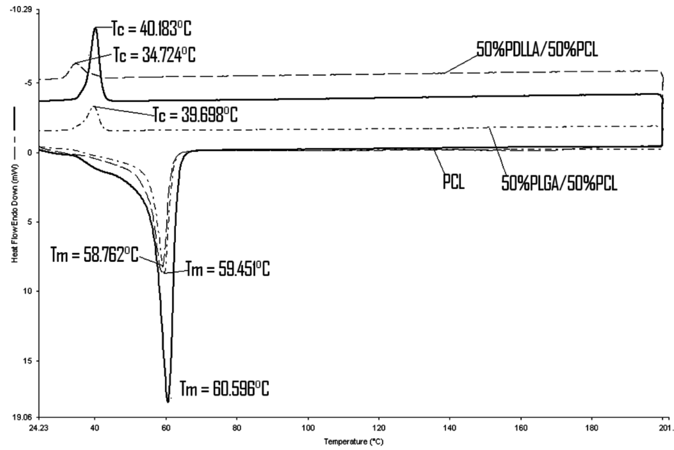


FIG. 4. DSC thermographs of compression moulded PCL homopolymer, and 50%PDLLA/50%PCL and 50%PLGA/50%PCL blends.

chemical structure[20]. Similar work carried out by Rahman and Mathowitz (2004) found evidence of miscibility between different grades of PLGA and PLA.[21] In the 20%PDLLA/80%PLGA blend, there is a shift in the $T_m$ value and a decrease in peak area when compared to the DSC thermograms of unblended PLGA.

The $T_m$ of PLGA is not visible in the 50% and 80% blends of PDLLA with PLGA. This change in the $T_m$ of PLGA correlates with work carried out by Kang et al. (2008) whom found there was a reduction in crystallinity in PLGA/PLA blends (with increasing PLA content) when blended as microparticles using solution-enhanced dispersion by supercritical fluids[22]. DSC thermographs of compression moulded PCL homopolymer, and 50%PDLLA/50%PCL and 50%PLGA/50%PCL blends are shown in Figure 4. PCL was found to have a $T_m$ of approximately 60°C in agreement with the literature[23]. The blends also exhibited only one endothermic peak, which is most likely related to the $T_m$ of PCL, and this broad melting peak masked the $T_g$s of the PDLLA and PLGA. This rendered any attempt to evaluate miscibility, by changes in $T_g$ of the polymers inconclusive. Similar masking behaviour was found for blends of PDLLA and PCL in DSC studies carried out by Yang et al.[24]

DSC can also be used to evaluate the percentage crystallinity of a polymer or polymer blend. Crystallinity influences many of the polymer properties, such as, hardness, modulus, tensile properties, stiffness, melting point and degradation. In this study, the percentage crystallinity was calculated using data obtained from the DSC thermograms and equations summarised by Sperling[25]. The percentage crystallinity of PCL is one of the most important properties controlling the degradation rate and the physical strength of the polymer[26]. The crystalline phase provides biodegradable polymers with the necessary mechanical strength for use in medical devices[27]. The percentage crystallinity calculated for compression moulded PCL homopolymer was 53.5%. Regularity in its molecular structure renders Poly (ε-caprolactone) highly crystallisable and in its usual state it is approximately 50% crystalline depending on molecular weight and processing conditions[28,29]. The number average molecular weight quoted by Dow for this grade of PCL was approximately 50,000.

Figure 4 demonstrates the effect of blending PCL with PLGA and PCL with PDLLA (blend ratio 50:50) on the crystalline peak. In work carried out by Tsuji and Ikada (1996) it was found that PCL can crystallise in the presence of other aliphatic polymers. In this work, the calculated crystallinity decreased from 44.4% to 5% as the blend ratio of PDLLA with PCL increased from 20:80 to 80:20[30]. There was a similar decrease in the calculated percentage crystallinity in PLGA/PCL samples, based on the same blend ratios. These results were calculated based on the presumption that PCL alone contributed to the melting point. The DSC thermographs of the blends show a reduction in area of the $T_m$ peak when compared to the PCL homopolymer, which coincides with the change in crystallinity. Previous work carried out by Tsuji and Ikada (1996) indicate that within blends of PDLLA and PCL, the disordering of the spherulite structure of PCL could be attributed to the addition of small amounts of PDLLA [30]. Therefore based on the results obtained, percentage crystallinity can be controlled by blending of PCL with PLGA or PDLLA, which in turn may be used as a means of controlling the degradation rate of the polymers.

## Phosphate Buffer Solution Adsorption

Phosphate buffer solution (PBS) adsorption studies were carried out on thin film samples of PLGA, PCL, PDLLA and blends of these polymers. The objective of these studies was to evaluate the adsorption properties of the polymers

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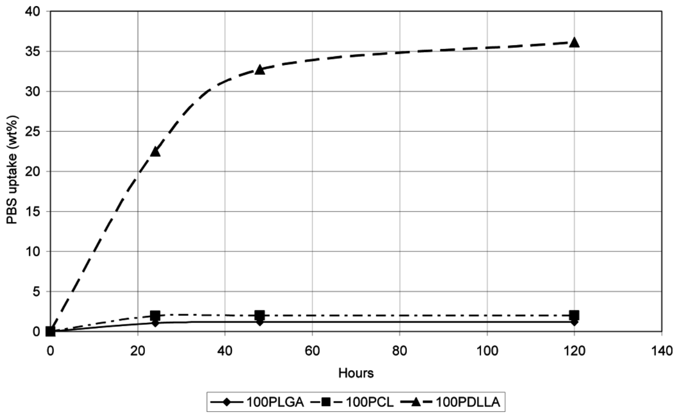


FIG. 5. PBS adsorption of PDLLA, PLGA and PCL over a 120-hour period.

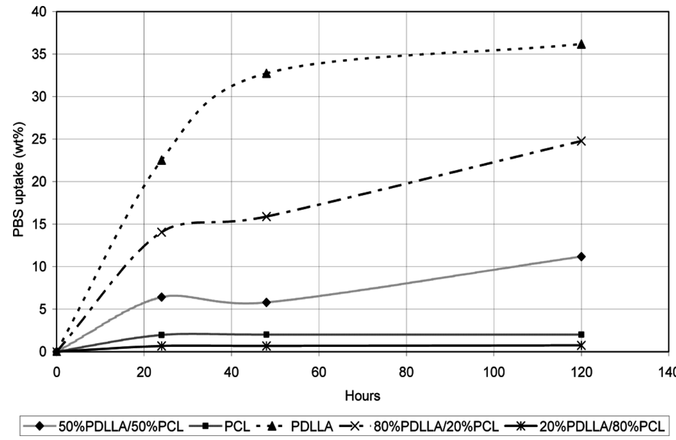


FIG. 6. PBS adsorption of PDLLA, PCL and PDLLA/PCL blends.

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and blends. Adsorption analysis provides important information relating to the physical properties and biodegradability of polymers/polymer blends. For this examination, the samples were immersed in PBS with a pH of 7.0 for 120 hours and the PBS adsorption percentage was calculated at regular time intervals, as illustrated in Figure 5 for PDLLA, PLGA and PCL. It can be clearly seen that the PDLLA films exhibited the highest PBS uptake. This finding is comparable with work carried out by Tarvainen et al. (2006), whom observed a water adsorption of greater than 50 wt% after 7 days for solvent cast PDLLA polymer films[31]. Degradation of aliphatic polyesters occurs faster in amorphous domains than in crystalline domains, as water penetration is easier within a disordered network of polymer chains[32]. The PDLLA used in this study is an amorphous polymer as demonstrated by the absence of a $T_m$ (Figure 2), and this may account for the high PBS uptake. Films of PCL were found to have a PBS uptake of approximately 2 wt% after 120 hr, which is in agreement with work carried by Zhang et al. (1995) under similar test conditions[33]. The percentage PBS uptake of the PLGA films proved similar to work carried out by Loo et al. (2005), with the films exhibiting a PBS uptake of less than 2 wt% after 120 hr[34].

Diffusion through a polymer occurs by the small molecules passing through voids and gaps between the polymer molecules. The size of the gaps between the polymer molecules depends on the physical state of the polymer, i.e., whether it is glassy, rubbery or crystalline. The molecules of semicrystalline polymers in the crystalline regions are packed together. This packing of the molecules lowers the amount of gaps or voids available for diffusion. In this study, the polymers PCL and PLGA are both semi-crystalline, which mainly accounts for their low percentage PBS uptakes. The difference in the $T_g$s of these polymers may have played a role in the minor difference in PBS uptake between the two samples. The $T_g$ of PCL is $-60°C$[23], which allows the polymer chains a higher degree of mobility at room temperature[35], whilst PLGA has a $T_g$ of 65°C (Figure 2). Generally, water uptake is faster in amorphous domains above the $T_g$ than below the $T_g$ because of the enhanced mobility of chain segments[32].

The PBS adsorption of thin film blends of PDLLA/PCL is illustrated in Figure 6. There is a considerably higher level of percentage PBS uptake in the 50%PDLLA/50%PCL and 80%PDLLA/20%PCL blends, than in the PCL and the 20%PDLLA/80%PCL samples. The increase in level of PBS uptake in the PCL/PDLLA blends at higher PDLLA contents is most likely a consequence of the reduction in crystallinity, as detailed in the previous section, thus the higher percentage of amorphous polymer present in the samples is likely to be responsible for this trend. A possible reason for the lower PBS uptake in the 20%PDLLA/80%PCL blend may be the enclosure of the amorphous PDLLA phases by the semi-crystalline polymer PCL as described by Tsuji et al.[30]

The PBS adsorptions of the thin film blends of PLGA/PCL were also analysed. Blends with the greatest levels of PCL incorporation exhibited the highest PBS uptakes. However, the overall difference in PBS uptake was minimal with percent adsorption between 1 and 2.5% for all of the blends and base polymers. Finally, the PBS adsorption of thin film PDLLA/PLGA blends was examined as shown in Figure 7. As with the PDLLA/PCL blend films, there was a similar pattern in that the PBS percentage uptake increased as the blend ratio of PDLLA increased in the blend. These differences in PBS uptake are again related to the difference in morphology of the polymers, as PCL and PLGA are semi-crystalline, while PDLLA is an amorphous polymer.

## Mechanical Properties of Thin Film Polymer Blends

The mechanical properties of biodegradable polymers are extremely important in medical device applications

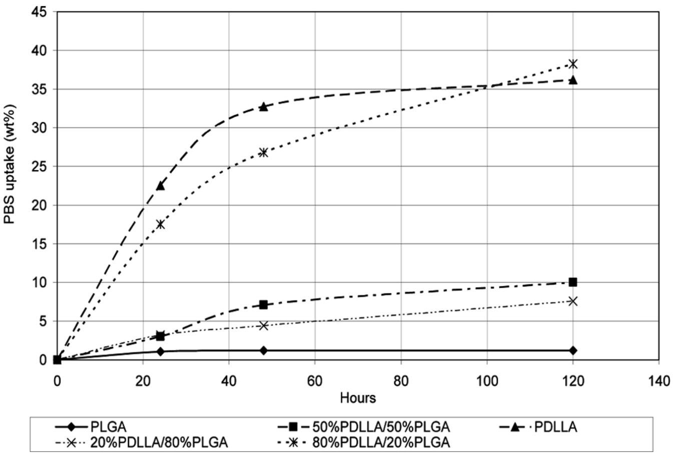


FIG. 7. PBS adsorption of PLGA, PDLLA and PLGA/PDLLA blends.

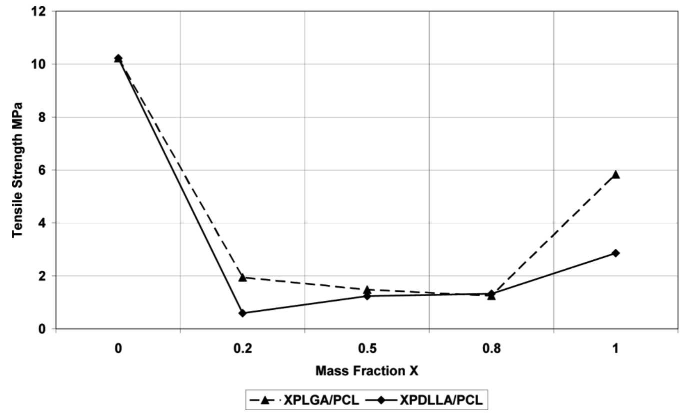


FIG. 8. Plot of tensile strength across the composition range for PDLLA/PCL and PLGA/PCL blends.

such as implantable macro devices and surgical sutures because these objects are often subject to considerable physical stress[27]. In order for polymers to be useful for diverse applications, it is necessary to be able to adjust the material properties to satisfy engineering requirements. Blending is a well-established strategy for tuning material properties and can radically alter resultant material properties, which depend sensitively on the mechanical properties of the base components. In this study, the mechanical properties of the polymers and the polymer blends were assessed using tensile testing. Tensile testing is one of the most widely used analytical techniques for assessing the mechanical properties of biodegradable polymers[28,30,36].

Table 2 shows tensile strength and Young's modulus values of thin polymer films of PLGA, PDLLA and PCL. The Young's modulus of 0.277 Gpa for PCL is close to the value of 0.282 Gpa, which was obtained by Vertuccio et al. (2009) for compression moulded films of PCL[37]. In Figures 8 and 9, the tensile strength and Young's modulus of PLGA/PCL and PDLLA/PCL are plotted across the composition range. The mechanical properties of blends of PLGA/PCL and PDLLA/PCL across all blend compositions were lower than the unblended polymer films. The mechanical properties of blends of PDLLA or PLLA with PCL or PEO have been previously shown to be greatly affected by altering the polymer mixing ratio[30]. The results of the mechanical analysis provide evidence to suggest that the blends of PDLLA/PCL and PLGA/PCL are immiscible. The miscibility of polymers plays a strong role in the mechanical properties of the polymer blends. Mixing two polymers usually leads to immiscible blends, characterised by a coarse, metastable morphology and poor adhesion between the phases. The properties most affected because of this phase-separated morphology are elongation-at-break, toughness and tensile strength[28]. A possible factor that likely contributed to the reduction in strength of the polymer blends is the change in the crystallinity resulting from the addition of either PDLLA or PLGA, as highlighted.

Presented in Figures 10 and 11, the tensile strength and Young's Modulus of blends of PDLLA/PLGA are plotted across the entire composition range. There is some evidence of miscibility between the blends, as the sample of 50%PDLLA/50%PLGA has a Young's Modulus greater than values obtained for PLGA or PDLLA, and the

TABLE 2
Tensile strength and Young's modulus of PDLLA, PLGA and PCL

| Sample | Tensile strength (Mpa) | Young's Modulus (Gpa) |
|---|---|---|
| PDLA | 2.85 | 0.163 |
| PLGA | 5.82 | 0.193 |
| PCL | 10.22 | 0.277 |

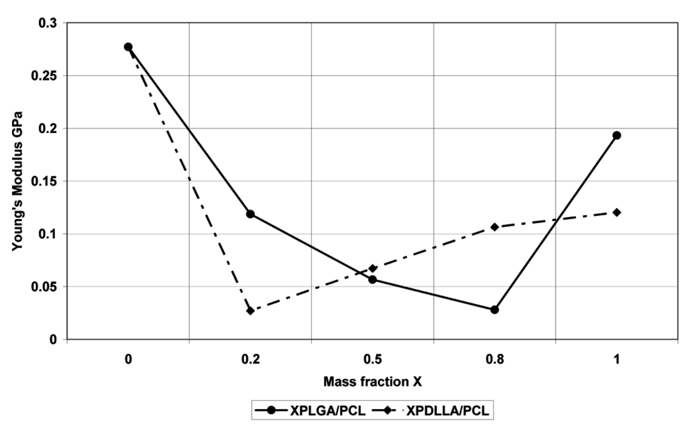


FIG. 9. Plot of Young's Modulus across the composition range for PDLLA/PCL and PLGA/PCL blends.

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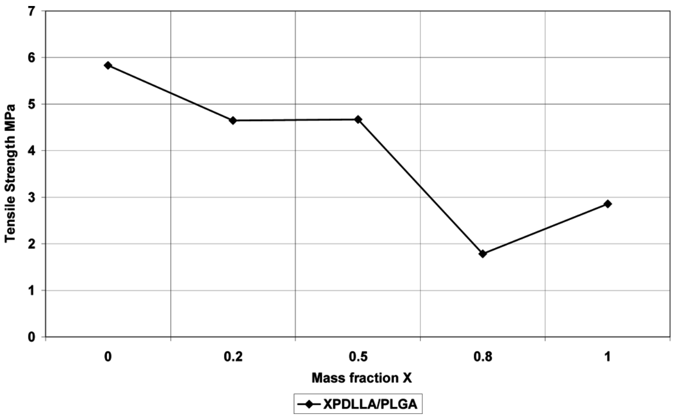


FIG. 10. Plot of tensile strength across the composition range for PDLLA/PLGA blends.

corresponding tensile strength is greater than the values obtained for PDLLA. The 20%PDLLA/80%PLGA blend has a higher tensile strength than the PDLLA but lower than PLGA. Rahman and Mathowitz (2004) have also presented work which provides evidence of miscibility between different grades of PLGA and PLA[21].

The changes in tensile properties of the thin film PDLLA/PLGA blends could also be related to dense chain packing in the amorphous regions due to a strong interaction between the l- and d-unit sequences, which is dependent on the polymer-mixing ratio[30]. The 80%PDLLA/20%PLGA blend shows a decrease in tensile strength when compared to the PLGA or PDLLA. This would indicate that at certain blend ratios, PDLLA/PLGA may not be entirely compatible. Also some miscible polymer blends of PLA and their copolymers with other polymers have become immiscible as the molecular weight increased because of the decreased polarity of polymer chains and the entropy of mixing[38].

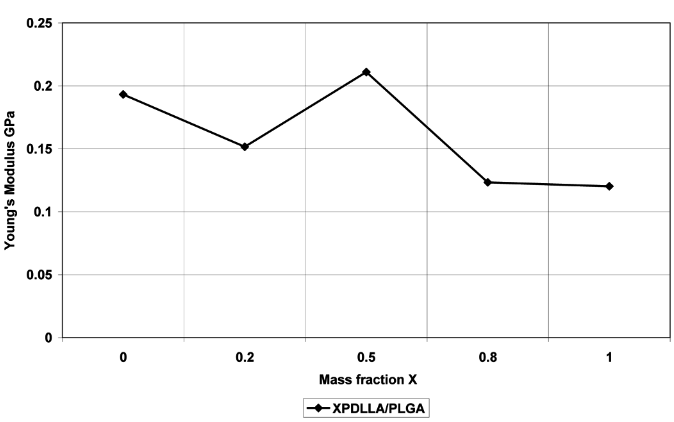


FIG. 11. Plot of Young's modulus across the composition range for PDLLA/PLGA blends.

## Surface Analysis of Thin Film Polymer Blends

Polymer surfaces and surface properties are important, and can often be directly related to bulk properties in polymeric materials. Hence, any surface property, i.e., surface energy, is expected to be influenced by changes in the bulk, including increases/decreases in the extent of phase separation and the degree of compatibility[39]. Moreover, surface analysis techniques are commonly used to assess the suitability of polymer blends for cell adhesion or drug elution[40]. In this study, contact angle analysis was chosen as the method of choice for investigating the sample surfaces. The contact angle value recorded for PCL was 77°, which is within the range 74–78° reported by Tiaw et al. (2005) for PCL membranes[41]. A contact angle of approximately 76° was obtained for PDLLA. Reported values of water contact angle for PDLLA without surface modification ranges from 69–87°[42]. PLGA was found to have a contact angle of approximately 72°, which is similar to the values of 68–77° for untreated PLGA films cited by Khang et al.[43] Figure 12 illustrates the contact angle values recorded for blends of PLGA/PCL. It was observed that the base polymers had slightly higher contact angles than the blends, a trend which was also apparent for the PDLLA/PCL and PDLLA/ PLGA samples. However, in all cases such changes in contact angles were within a narrow range. Other authors have reported that by changing the relative base polymer proportions the domain structure and surface morphology can be changed[44,45]. This is particularly the case for immiscible blends as there is a tendency for one of the base polymers to be enriched at the surface with regards to the other[44,45]. Gogolewski and Pennings reported that a contact angle within the range 60–80° is suitable for the surface attachment and growth of cells[46]. All of the recorded contact angles were within this quoted range and thus have potential for such applications.

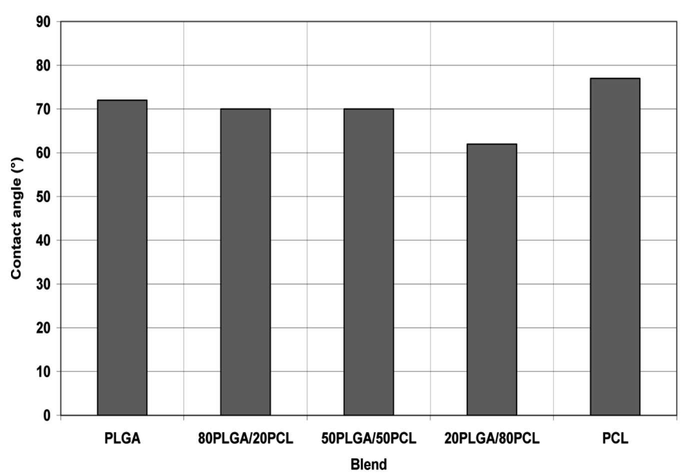


FIG. 12. Contact angles of PLGA, PCL and PLGA/PCL blends.

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## CONCLUSION

A variety of blends based on PDLLA, PLGA and PCL were fabricated using both solvent casting and compression molding techniques. When the feed ratio of PDLLA or PLGA exceeded 20 wt% in the blends with PCL, undulations formed on the surface of the solvent cast samples. Therefore, compression moulding was chosen for processing of the samples, as films with fine aesthetic appearance were produced following a premixing step using this technique. The DSC analysis provided evidence of miscibility and interaction between the PDLLA/PLGA blends. It was not possible to assess the miscibility of PDLLA/PCL and PLGA/PCL blends using the glass transition temperature criteria, due to the masking of the relevant $T_g$ peaks of the base polymers by the $T_m$ of the homopolymer.

Thin films of PCL exhibited the highest tensile strength, while blends of PCL with PLGA or PDLLA decreased in strength when blended together, indicating their immiscibility. Blending of PCL with PLGA or PDLLA had an obvious effect on the percentage crystallinity of the samples, which in turn may be used as a means of controlling the degradation rate of the polymers. This hypothesis is strengthened as PBS adsorption was highly dependent on the crystallinity of the blends. In summary, the results indicate the feasibility of processing blends with a diverse range of properties, making them attractive for potential biomedical applications.

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